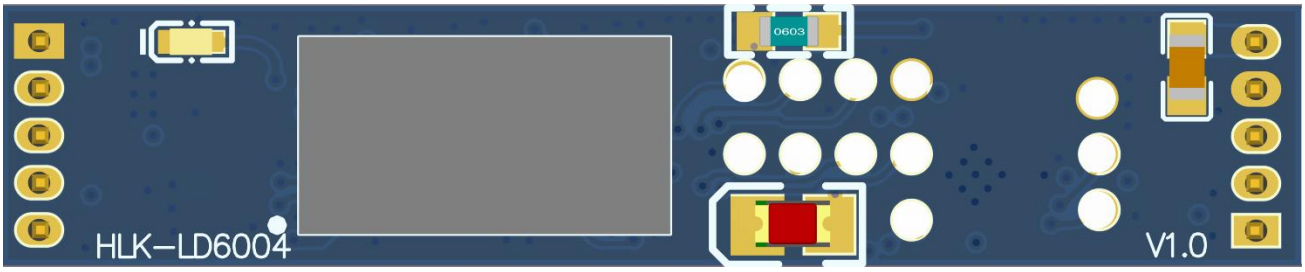
Shenzhen Hi-Link Electronic Co., Ltd

HLK-LD6004

3D Presence Radar Module Technical Specification Sheet



Shenzhen Hi-Link Electronic Co., Ltd
2025.9

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1. Product profile

HLK-LD6004 is a radar sensing module developed by Hi-Link based on the ADT6101P chip. It integrates a 57-64GHz RF transceiver system, a 2T2R AIP packaged antenna, 1MB flash, a radar signal processing unit, and an ARM® Cortex®-M3 core. This module operates based on the FMCW signal processing mechanism and combines radar signal processing algorithms to achieve high-sensitivity human body state sensing. It can identify movement and micro-motion, as well as stationary human bodies, and can calculate auxiliary information such as target distance and speed.

This module is in a two-transmit-two-receive antenna form, with a relatively wide beam angle, and is suitable for ceiling installation. It supports GPIO and UART signal output, and can be flexibly applied to different intelligent scenarios and products.

2. Product characteristics

- Radar detection based on FMCW frequency-modulated continuous wave signal
- Wide detection angle, supporting horizontal ±60° and pitch ±60° detection
- Supports side-mounted mode switching, with a maximum sensing distance of 6m
- Maximum support for 3-person target motion tracking, outputting x, y, z coordinates of the personnel position.
- Capable of realizing indoor personnel motion and stillness detection, accurately identifying within the range, supporting division of sensing range, and shielding interference inside and outside the range
- General UART interface, providing communication protocol
- Supports UART parameter adjustment to meet different scene requirements
- Compact in size, only 25*31.5mm, supporting pin header connection and surface mount connection methods
- Not affected by temperature, humidity, noise, airflow, dust, light, etc. environmental factors

3. Application scenarios

✧ Smart Home Appliance Application

Based on the detection of whether there is a person in the room, the working mode of home appliances (working, standby, off) is adjusted in real time to achieve the intelligence of home appliances.

✧ Human Detection Light Application

Performs sensing on whether there is a human in the current space and automatically controls the light switch. For example, public area lighting, office lighting, and various sensing lights.

✧ Smart Home Scene Application

For places such as homes, hotels, offices, and bathrooms, it is necessary to conduct real-time detection of whether there is a person entering or present within the place. Then, it can achieve security, electrical appliance control, and personnel monitoring methods. It can be combined with relevant IoT support platforms to achieve effective application in related places.

4. Electrical characteristics and parameters

4.1Functional parameters

Parameter content	Min	Typical	Max	unit	Remarks
Detection target			3		
Distance accuracy		0.4		m	
Side-mounted sensing distance	0		6	m	
Recognition accuracy rate		95		%	

4.2 Electrical characteristics

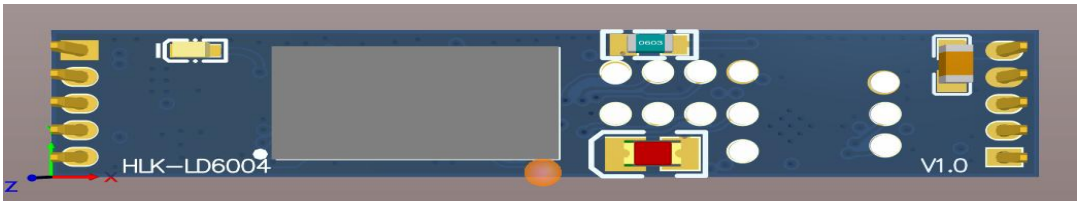
Operational parameter	Min	Typical	Max	unit
Operating voltage (VCC)	3.1	3.3	3.5	V
Operating current (ICC)		135	600	mA
Operating temperature (TOP)	-20		85	℃
Storage temperature (TST)	-40		85	℃

4.3 RF characteristics

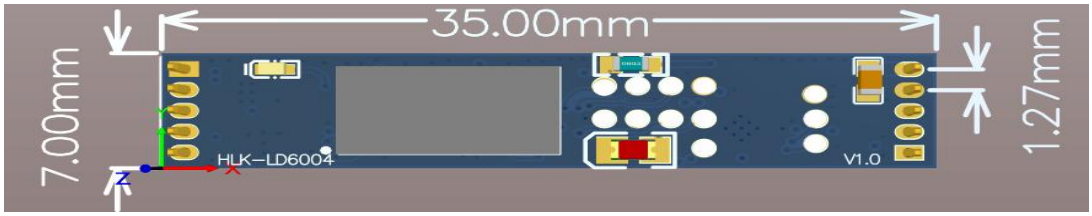
Operational parameter	Min	Typical	Max	unit
Operational frequency	58		64	GHZ
Emission Power (Pout)		12		dBm
Antenna gain		4		dBi
Horizontal beam (-3dB)	-60		60	°
Vertical beam (-3dB)	-60		60	°

5. Hardware description

5.1Overall dimensions



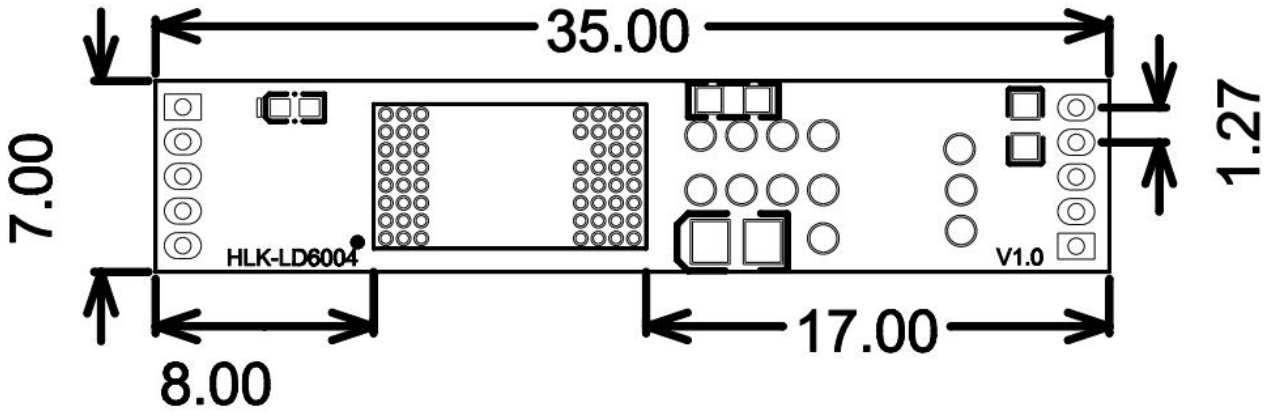
Front view of the module



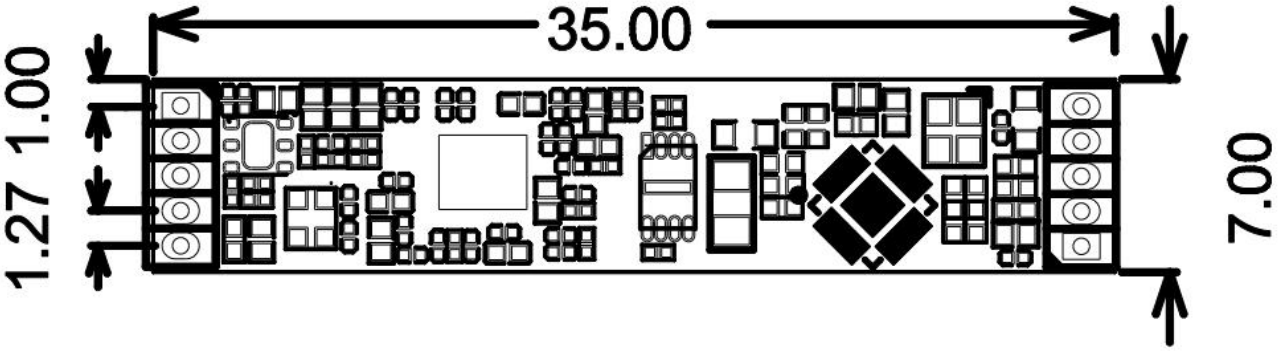
Back view of the module



Module pin diagram



TOP View

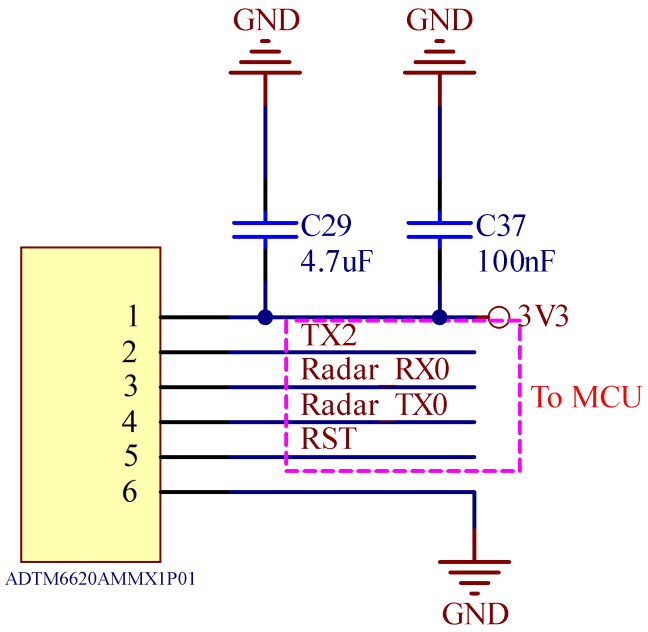


BOTTOM View

5.2 Pin definition

Pin number	Pin name	Description	Remarks
1	3.3V	POWER_INPUT 3.3V	
2	TX2	GPIO_P20/uart2 txd	If there is someone within the area, output a high level.
3	RX0	GPIO_P01/uart0 _txd	
4	TX0	GPIO_P00/uart0 _txd	
5	RST	Reset Signal input	Built-in pull-up, effective at low levels
6	GND	GND	
7	3.3V	POWER_INPUT 3.3V	
8	SWCLK	SWD Debug Clock	Debugging usage
9	SWDIO	SWD Debug Signal	Debugging usage
10	GND	GND	

5.3 Module peripheral reference design



5.4 Boot configuration

	BOOT1	BOOT0	Remarks
Configuration level	0	1	Flash starts within the module
Pipe foot position number	Pin8	Pin12	

* BOOT 1, BOOT 0 module internal are pulled up. The BOOT 1 must be connected to a low level before the module starts

6. Usage and Configuration

6.1 Typical application circuit

The HLK-LD6004 module can directly use the TX2 pin output to detect the target information (with a high level indicating presence and a low level indicating absence). At the same time, UART0 outputs the detection results according to the specified protocol. The serial port data includes the target position, speed auxiliary information, and users can flexibly use it according to specific application scenarios.

The module is powered by 3.3V, and the input power supply capacity requirement is greater than 1A.

The output voltage of the module's IO ports is 3.3V. The default baud rate of the serial port is 115200, and there is no parity check.

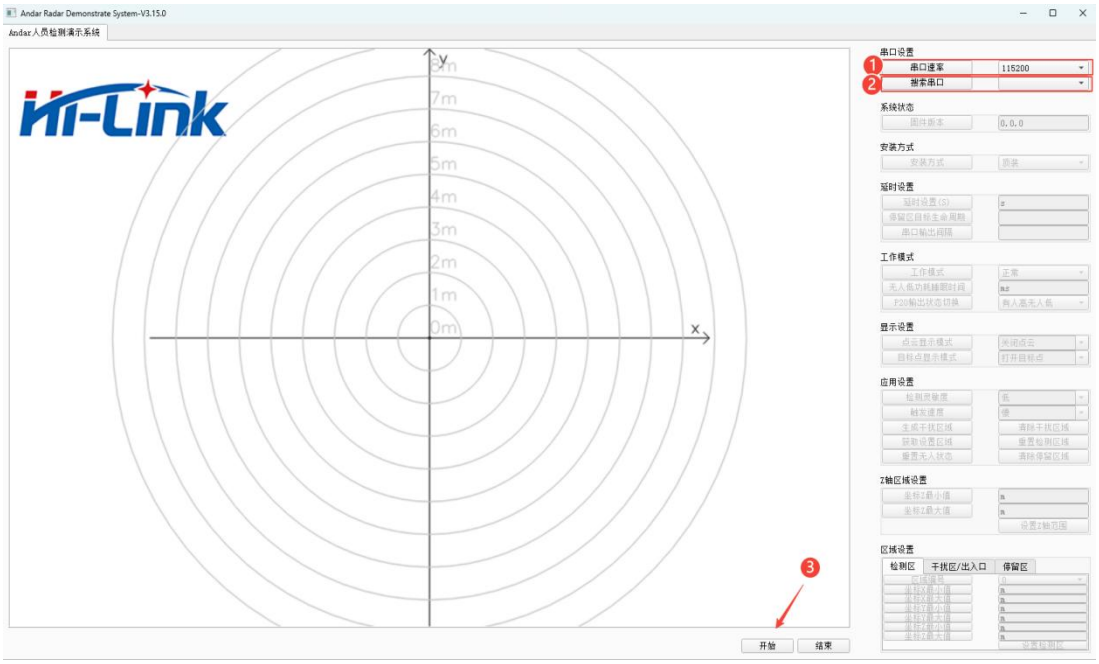
6.2 GUI visualization tool application

1. Device connection

1) Set the Baud rate to 1,15,200

2) Select the connected serial port

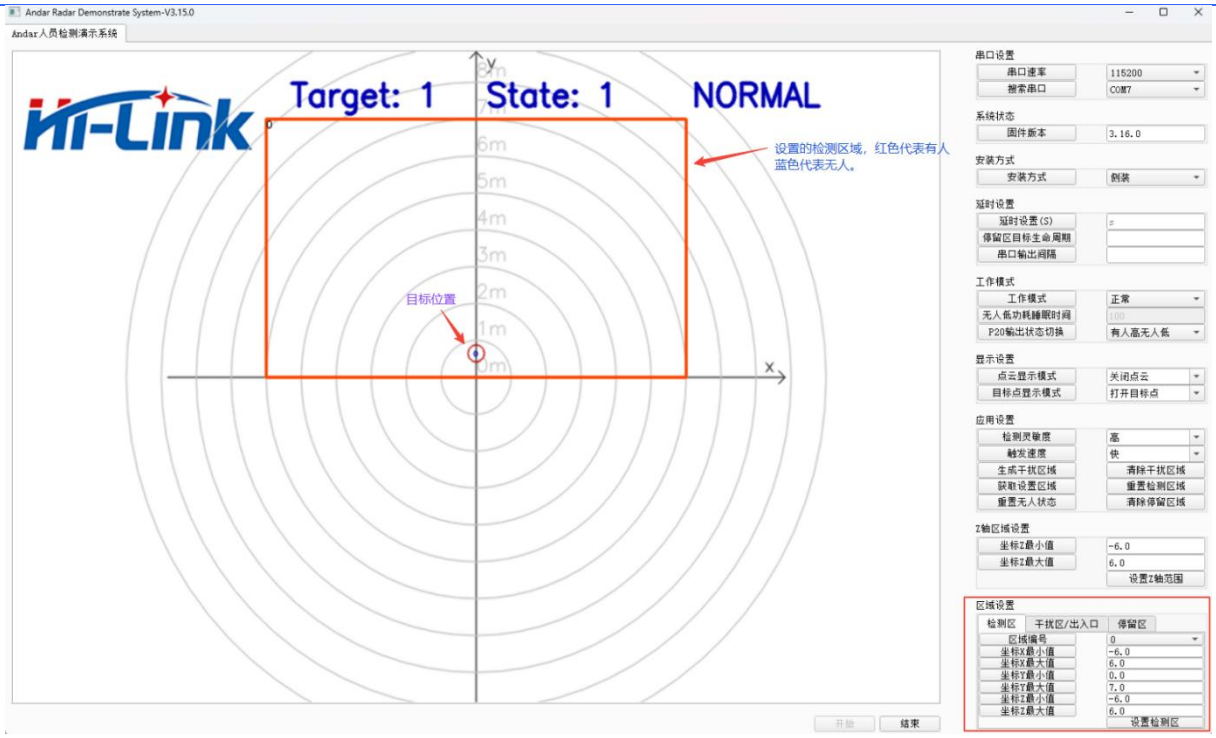
3) Click the [Start] button, and the module will start the detection



2. **Set the detection area**

1) Set the coordinates of the detection area (with the module as the origin). A maximum of 4 detection areas can be set. If there is a ceiling-mounted air conditioner installed in the room, the interference from the air conditioner can be filtered by limiting the z-axis range.

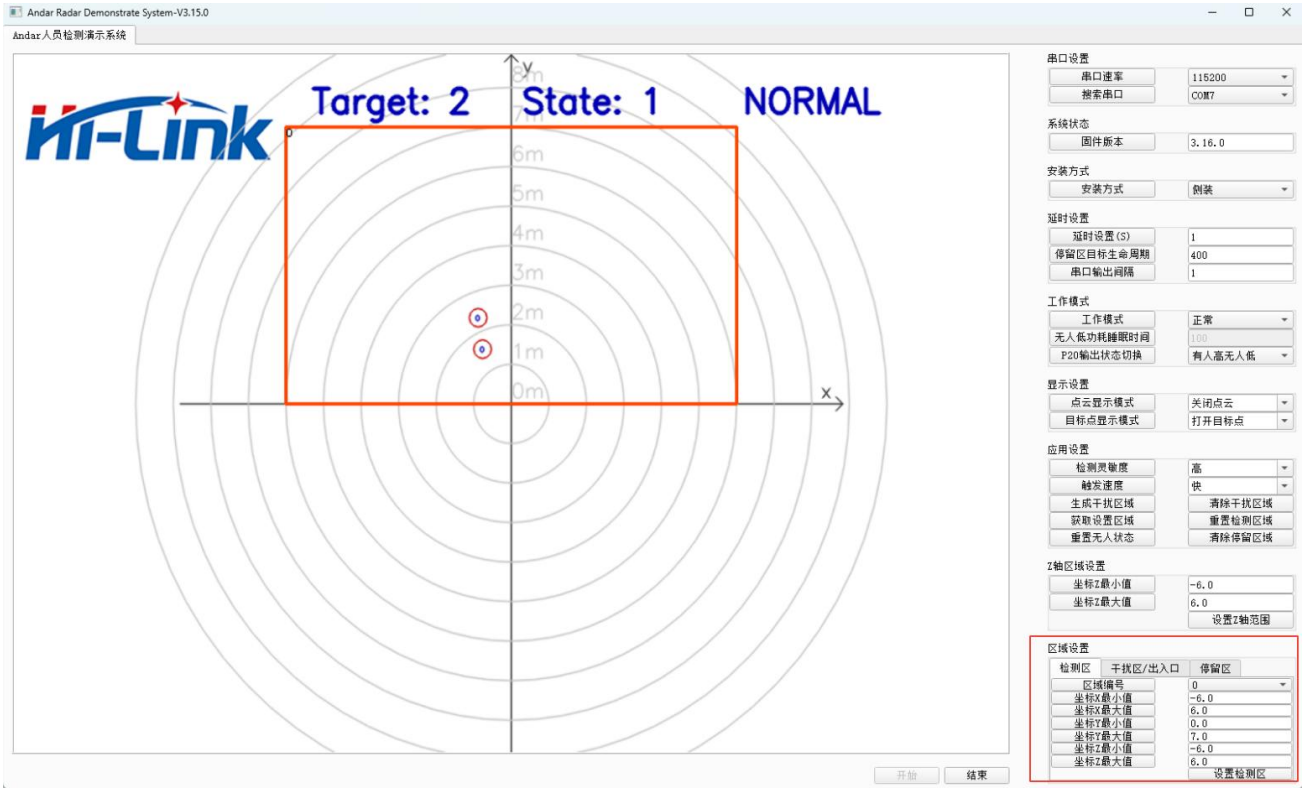
2) Click the "Set Detection Area" button. After setting the detection area, the modules outside the detection area can still detect, but only when triggered within the detection area will the TX2 output a high level. Power loss saves the data.



3. **Set the interference area**

1) Set the coordinates of the interference area (based on the actual usage scenario, some areas that may cause interference to the radar sensor, such as air conditioners, fans, curtains, etc., should be shielded). Targets within this area will not trigger the sensor, that is, TX2 will not output a high level. Up to 4 interference areas can be set.

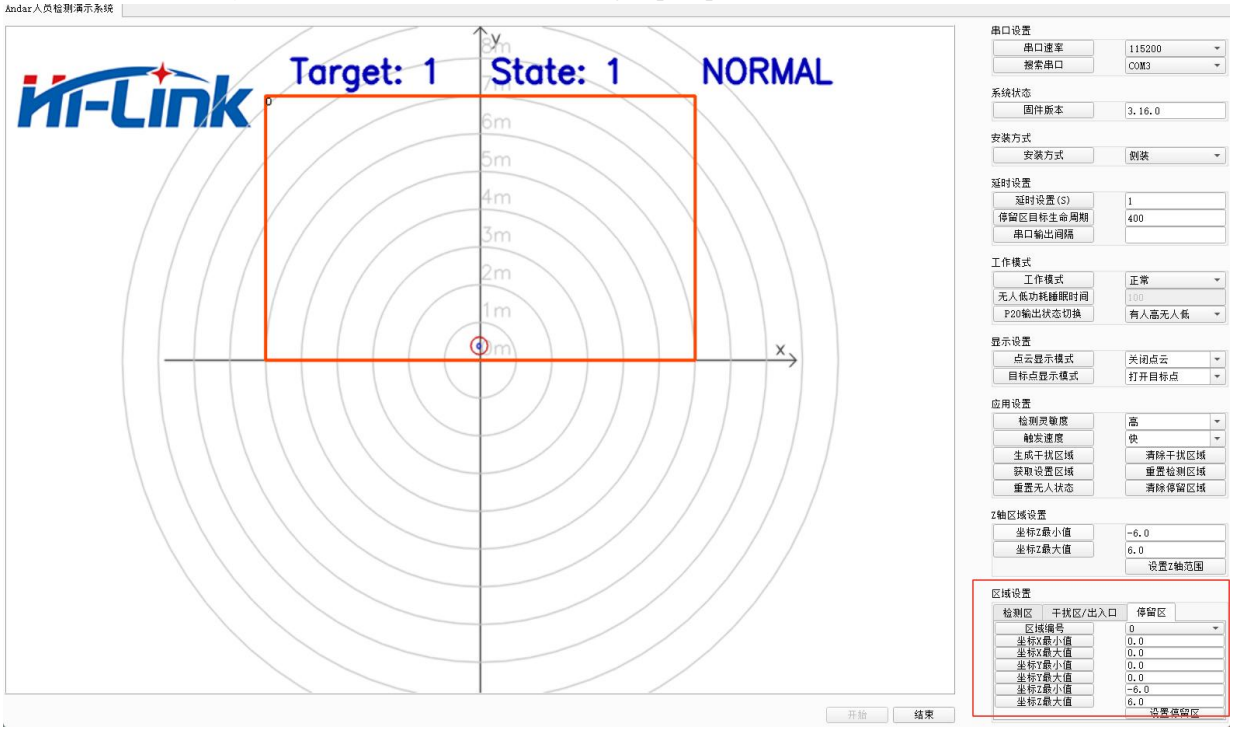
2) Click the "Set Interference Area" button, and save the settings upon power loss.



4. Set the stay area

1) Set the coordinates of the stay area (for some scenarios with higher requirements, such as deep sleep mode or the bathroom glass partition scenario where the state needs to be maintained well and the target should quickly become unoccupied after leaving, the delay for the stay area and the detection area can be set separately).

2) Click the "Set Stay Area" button and save the settings upon power loss.



5. **Other functions introduction**

1) [Software version]: Connect the GUI to automatically identify the software version number

2) [Installation Method]: This module supports switching between top-side and side-side installation methods.

3) [Delay Setting]: When someone is detected, the duration for maintaining the presence status.

4) [Target Lifecycle in Staying Area]: The delay time for the staying area and the detection area are independently set. This is mainly for areas with relatively weak signals (such as bathroom glass partitions, beds). Within the set delay time, there will still be a certain number of point clouds generated by the target, allowing the target to remain. When the target exits the staying area and then exits the detection area, the time for switching from "person" to "unperson" is still based on the set delay time. For example, if the delay setting time is 10 seconds, the staying area target's delay time is 30 seconds, and when someone exits the staying area and then exits the detection area, the time for switching from "person" to "unperson" is 10 seconds. The setting lifecycle = time x 20.

5) [Serial Port Output Interval]: Set the radar reporting interval. The default is 50ms per frame. Set it to 10, then it is 500ms per frame.

6) [Low Power Mode]: When no one is present, it switches to 1T1R mode. After detecting movement, it switches to 2T2R mode to detect stationary targets. When enabled, the average working current for no one is 2.5mA. When not enabled, it is always 2T2R mode, with an average working current of 135mA. Enabling the low power mode allows adjusting the sleep time. The sleep time is not recommended to exceed 500ms.

7) [Turn off Radar P20 Output High Mode]: Turn off the radar. The TX2 pin continuously outputs a high level.

8) [Turn off Radar P20 Output Low Mode]: Turn off the radar. The TX2 pin continuously outputs a low level.

9) [Strong Reflection Mode]: Use for environments with a lot of metals.

10) [Switching of P20 Output State]: The default is "high for person, low for no one". When there is someone, the TX2 outputs a high level, and when there is no one, it outputs a low level. "Low action (500ms low)" when there is someone, the TX2 outputs a 500ms low level; "High action (500ms high)" when there is someone, the TX2 outputs a 500ms high level.

11) [Point Cloud Display Mode]: Enable or disable the display of the original point cloud data.

12) [Target Display Mode]: Enable or disable the display of target points.

13) [Detection Sensitivity]: Set the sensitivity level. Low sensitivity only detects movement.

14) [Trigger Speed]: The time when someone is detected and triggers. Fast, medium, and slow are 0.5s, 1s, and 5s respectively.

15) [Generate Interference Area]: After installation, when there is no one in the detection environment, turn on the interference source in the room. When the radar detects the interference target, click [Generate Interference Area] to perform point-based shielding on the interference target points.

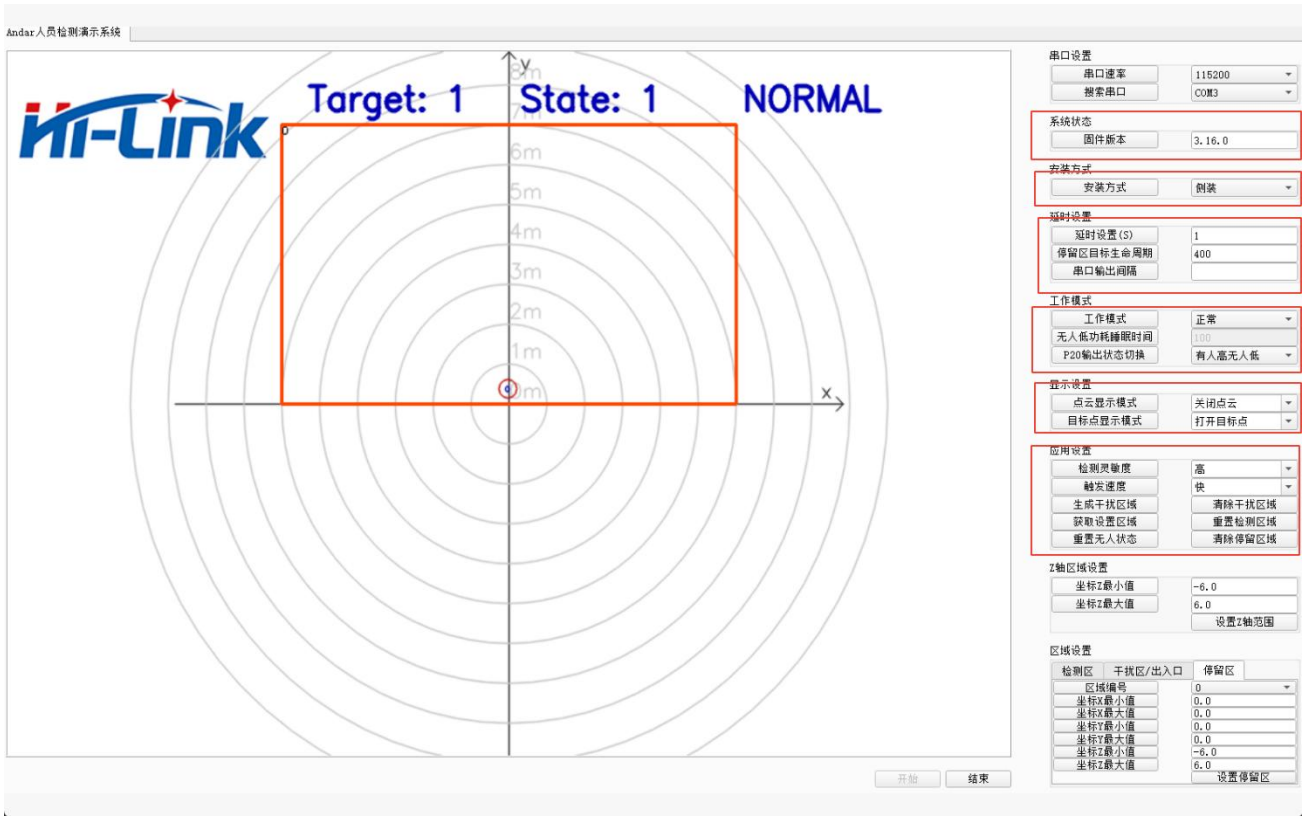
16) [Clear Interference Area]: Clear the interference area that was manually set or the interference area generated automatically.

17) [Get Set Detection Area]: The detection area and interference area set by the user are saved even when the power is off. Click this button to obtain the area previously set by the module.

18) [Reset Detection Area]: Restore the default 6*4m detection area.

19) [Reset Unperson State]: For scenarios with long delay times, it is convenient for testing. Force reset the detection state to "unperson".

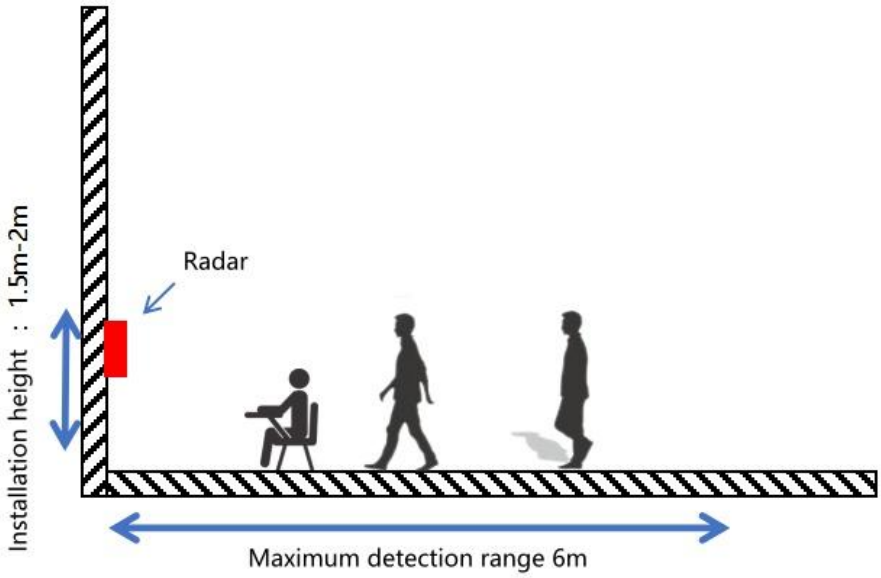
20) [Z-axis Area Setting]: Set the overall Z-axis detection range. The radar is the origin, meaning the minimum coordinate Z value is 0.



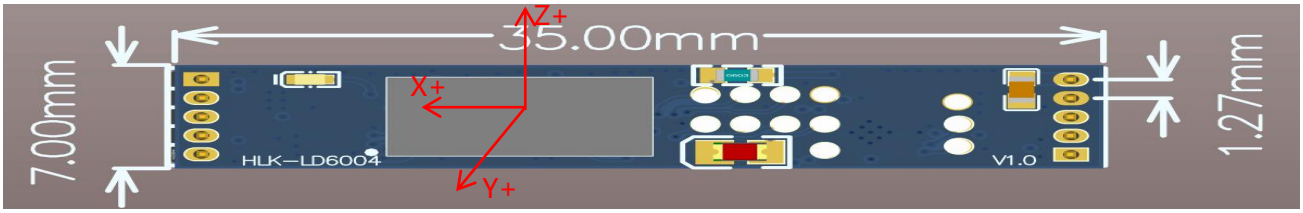
6.3 OTA upgrade

Refer to the 《OTA Upgrade Tool Manual _V1.0 》documentation

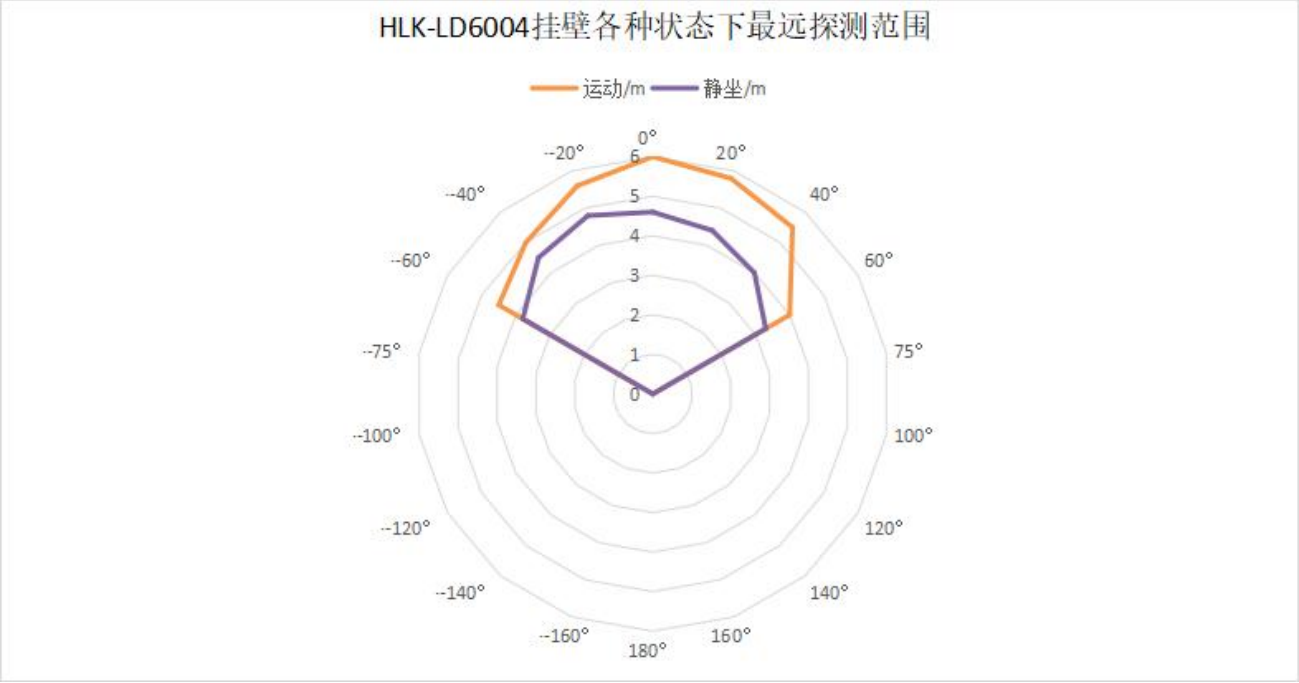
6.4 Installation mode and induction range



Side-mounted schematic diagram



Direction signs



Schematic diagram of side-mounted detection range

7. Precautions

1). The detection range of the radar module is highly correlated with the target's RCS and environmental factors. The effective detection range may vary with the environment and the target, so it is normal for the effective detection range to fluctuate within a certain range.

2). The radar module has extremely high requirements for power supply, requiring an input voltage of 3.1 to 3.5V, with a power supply ripple of ≤ 50mV and a current of ≥ 1A. If using a DCDC power supply, the switching frequency must be no less than 2MHZ.

8. Radome design

The radar antenna cover is used to protect the radar antenna from external environmental factors such as rain, sunlight, and wind. However, it has the following effects on the radar antenna: The dielectric loss and reflection loss brought by the antenna cover will reduce the effective power of the radar; It causes the antenna beam to distort, resulting in changes in the radar's effective coverage area; The reflection of electromagnetic waves by the shell reduces the isolation degree between the radar transmitting and receiving antennas and may cause the receiver to saturate; The phase of the electromagnetic waves changes when passing through the radar antenna cover, affecting the measurement of angles. Therefore, it is very necessary to design the radar antenna cover to reduce the influence of the shell and improve the radar performance.

Design requirement:

1. When selecting the material of radome, under the premise of ensuring firmness and low cost, the material with smaller dielectric constant and loss angle should be selected to reduce the influence of radome on radar performance.

Dielectric constant and dissipation factor of common materials are shown below:

material	dielectric constant () ε_r	Disguetion factor (tan δ)
Merlon	2.9	0.012
ABS	2.0-3.5	0.0050-0.019
PEEK	3.2	0.0048
PTFE （Teflon ®）	2	<0.0002

Plexiglass ®	2.6	0.009
glass	5.75	0.003
pottery and porcelain	9.8	0.0005
PE	2.3	0.0003
PBT	2.9-4.0	0.002

2. The radome is required to have a smooth surface and a uniform thickness
3. Radome antenna thickness design requirements

$$T = N \bullet \frac{c}{2f\sqrt{\varepsilon_r}}, \quad N=1, \ 2, \ 3\ldots$$

T: Radome thickness

c: light speed, 3×10^8 m/s;

f: center frequency

ε_r : Material permittivity, DK

4. Design requirements of radar antenna from the inner surface of the enclosure

$$d = N \bullet \frac{c}{2f} \quad N=1, \ 2, \ 3\ldots$$

3×10^8 And c: light speed, m/s;
f: center frequency
f=60GHz
c/2f=2.5mm

Revision History

Revision	Release Date	Description
V1.0	2025/09/25	First release



Shenzhen Hi-Link Electronic Co., Ltd

Address： 17F,Building E, Xinghe WORLD, Minzhi Street, Long Hua district, Shenzhen 51813

Email： sales@hlktech.com

Website： www.hlktech.net